

Determine whether the statement forms in 16-24 are logically equivalent. In each case, construct a truth table and include a sentence justifying your answer. Your sentence should show that you understand the meaning of logical equivalence.

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$

Let's start with the statement $p \wedge (q \vee r)$, which contains three separate statement variables.

Each variable has two possibilities: T or F.

$$2 \times 2 \times 2 = 2^3 = 8$$

p	q	r	$q \vee r$	$p \wedge (q \vee r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	T	F
F	T	F	T	F
F	F	T	T	F
F	F	F	F	F

Let's do the same thing for the statement $(p \wedge q) \vee (p \wedge r)$.

$(p \wedge q) \vee (p \wedge r)$ also contains three statement variables.

p	q	r	$(p \wedge q)$	$(p \wedge r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	F	F
F	T	F	F	F	F
F	F	T	F	F	F
F	F	F	F	F	F

Now we can examine the statement form by comparing the truth values from each corresponding truth table.

$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T
T	T
T	T
F	F
F	F
F	F
F	F
F	F

When examining the statement form we can now conclude that the statement form is logically equivalent.

Solution: Yes, the statement form is logically equivalent.